

# Pediatric Extracorporeal Cardiopulmonary Resuscitation: A Systematic Review

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**Objectives:** This systematic review aims to summarize the body of available literature on pediatric extracorporeal cardiopulmonary resuscitation in order to delineate current utilization, practices, and outcomes, while highlighting gaps in current knowledge.

**Data Sources:** PubMed, Embase, Scopus, Cochrane Library, and ClinicalTrials.gov databases.

**Study Selection:** We searched for peer-reviewed original research publications on pediatric extracorporeal cardiopulmonary resuscitation (patients < 18 yr old) and were inclusive of all publication years.

**Data Extraction:** Our systematic review used the structured Preferred Reporting Items for Systematic Reviews and Meta-Analyses methodology. Our initial literature search was performed on February 11, 2019, with an updated search performed on August 28, 2019. Three physician reviewers independently assessed the retrieved studies to determine inclusion in the systematic review synthesis. Using selected search terms, a total of 4,095 publications were retrieved, of which 96 were included in the final synthesis. Risk of bias in included studies was assessed using the Risk of Bias in Non-Randomized Studies of Interventions-I tool.

**Data Synthesis:** There were no randomized controlled trials of extracorporeal cardiopulmonary resuscitation use in pediatrics. A vast majority of pediatric extracorporeal cardiopulmonary resuscitation publications were single-center retrospective studies reporting outcomes after in-hospital cardiac arrest. Most pediatric extracorporeal cardiopulmonary resuscitation use in published literature is in cardiac patients. Survival to hospital discharge after extracorporeal cardiopulmonary resuscitation for pediatric in-hospital cardiac

arrest ranged from 8% to 80% in included studies, and there was an association with improved outcomes in cardiac patients. Thirty-one studies reported neurologic outcomes after extracorporeal cardiopulmonary resuscitation, of which only six were prospective follow-up studies. We summarize the available literature on: determination of candidacy, timing of activation of extracorporeal cardiopulmonary resuscitation, staffing/logistics, cannulation strategies, outcomes, and the use of simulation for training.

**Conclusions:** This review highlights gaps in our understanding of best practices for pediatric extracorporeal cardiopulmonary resuscitation. We summarize current studies available and provide a framework for the development of future studies. (*Pediatr Crit Care Med* 2020; XX:00–00)

**Key Words:** cardiopulmonary resuscitation; extracorporeal cardiopulmonary resuscitation; extracorporeal life support; extracorporeal membrane oxygenation; pediatric

## BACKGROUND AND HISTORICAL PERSPECTIVE

Although variations of portable cardiopulmonary bypass were reported in the literature as early as 1972 (1), the first cases of deployment of extracorporeal membrane oxygenation (ECMO) during cardiac arrest were reported in adults in 1976 (2). The first published report of ECMO use for cardiopulmonary resuscitation (CPR) in pediatrics was not until 1992 (3). Since then, pediatric extracorporeal CPR (ECPR) has increased exponentially (4). ECPR has been shown to be associated with improved survival compared with conventional CPR (CCPR) for pediatric in-hospital cardiac arrest (IHCA) (5, 6). The American Heart Association (AHA) first included consideration of ECPR in the 2005 Pediatric Advanced Life Support (PALS) guidelines (7) and has subsequently updated the 2015 guidelines to recommend a consideration of ECPR specifically in pediatric cardiac patients with IHCA (8) based on multiple studies demonstrating an association with survival benefits in this group (9–12). However, there remain significant gaps in our understanding of ECPR candidacy, timing, cannulation strategy, neurologic outcomes, and survival

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across the heterogeneous pediatric population. This systematic review aims to examine current available literature for practices and outcomes, as well as identify gaps in the literature.

## METHODOLOGY

This systematic review of peer-reviewed publications on pediatric ECPR in the medical literature followed recommendations in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (13). Using the PICO (Population, Intervention, Comparison, Outcome) framework, we formulated the study question: for pediatric patients less than 18 years old with cardiac arrest (P) who undergo ECPR(I), compared with patients who have not received ECPR (C), what are current resuscitation practices and outcomes(O)? We searched for studies on pediatric ECPR with the following inclusion and exclusion criteria.

### Inclusion Criteria

Original research publications on pediatric ECPR (< 18 yr) in peer-reviewed journals inclusive of all publication years

### Exclusion Criteria

- Non-English language publications
- Case reports
- Case series with less than 10 subjects, or studies in which the pediatric ECPR subgroup had less than 10 subjects
- Publications reporting on greater than or equal to 10 pediatric ECPR subjects as a subgroup of study population, but with no subgroup analysis of the pediatric ECPR subjects, such as presentation of survival or neurologic outcomes data
- Review articles
- Conference abstracts
- Letters to the editor
- Animal studies

### Review Process and Data Extraction

A list of search terms was created by physician reviewer (I.D.E.) and medical librarian (F.M.C.) and then subsequently agreed upon by two other physician reviewers (M.A.B., J.J.L.). Appropriate search strings were created using these terms. Using these criteria, a comprehensive query of five electronic databases was performed: PubMed, Embase, Scopus, Cochrane Library, and ClinicalTrials.gov. This initial search was done on February 11, 2019. **Supplemental File S1** (Supplemental Digital Content 1, <http://links.lww.com/PCC/B323>) details the search terms and search string formation used for the database searches. Identified publications were uploaded onto the web application Rayyan (Qatar Computing Research Institute, Doha, Qatar) for organization and blind reviewing (14).

The three physician reviewers (I.D.E., M.A.B., J.J.L.) reviewed the publications generated to determine which articles would be included in the systematic review synthesis. This process was done in three phases. During phase 1, the total number of publications was divided into three equal sections. Each physician

reviewer was then assigned a section and, working independently via Rayyan, reviewed articles in order to make determinations about inclusion or exclusion based on previously agreed-upon criteria. Abstracts were reviewed, and full-text articles were obtained and reviewed for any article where a determination could not be made by abstract alone. During phase 2, each of the three reviewers moved to a different section, identifying articles for inclusion. Reviewers' assignment of publications remained blinded through the first two phases, meaning that no reviewer could see the determinations of the preceding reviewer. After completion of phase 2, each publication had been independently reviewed by two physician reviewers. There emerged a list of articles with conflicting determinations (inclusion vs exclusion). In phase 3 of the review, reviewers' assignments were unblinded, and articles with conflicts were identified. For every article where there was a decision conflict between any two reviewers, the third reviewer assessed the article and determined inclusion or exclusion. Any disagreements after this stage were resolved by discussion.

The risk of bias of each study included in the synthesis was then assessed by I.D.E. using the Risk of Bias in Non-Randomized Studies of Interventions-I tool (15) at the outcomes level, and outcomes data were then extracted from each study. For each publication, the likelihood of bias in reported outcome measures was assessed based on study methods, definitions used, and format of results.

A repeat search of the databases using the same criteria was done on August 28, 2019, to check for new articles that had been published after February 11, 2019. Additional publications that met criteria were added to the synthesis after discussion by all three reviewers.

## RESULTS

The initial database search produced 6,758 publications, 3,870 after removal of duplicates. **Table 1** shows sources of publications retrieved. Each of the three reviewers assessed 1,290 publications during the first phase and then moved to another section of 1,290 publications during the second phase (blinded). After the second phase, there were 251 publications with conflicted assignments, which were resolved during the unblinded third phase. This yielded 94 included publications. The repeat search performed in August 2019 retrieved 225 additional articles. Only two of these articles met criteria and were added to the final list. Therefore, the final synthesis

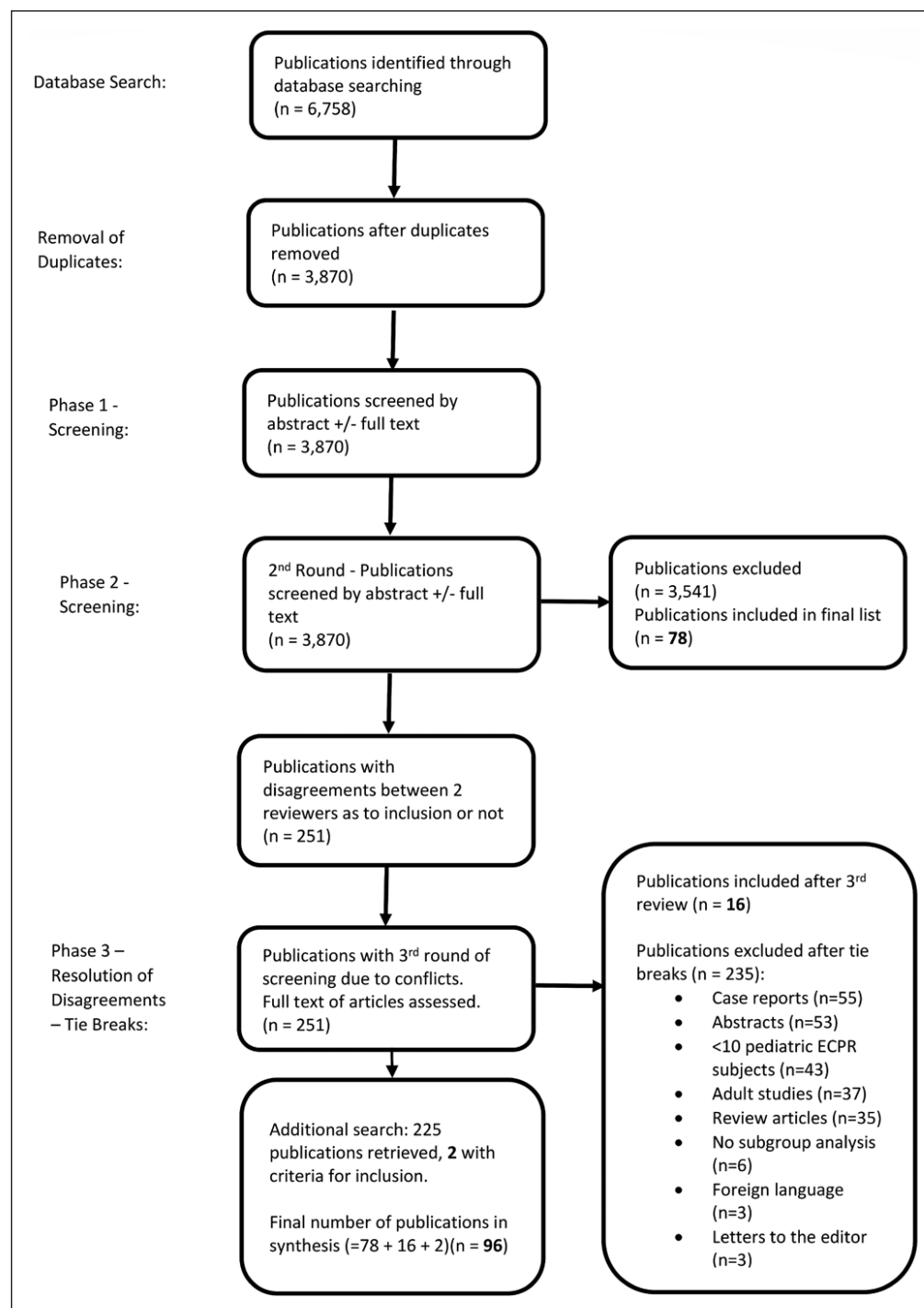
**TABLE 1. Data Sources of Publications in Initial Search**

| Database or Website | No. of Publications |
|---------------------|---------------------|
| PubMed              | 1,404               |
| Embase              | 3,263               |
| Scopus              | 1,987               |
| Cochrane library    | 74                  |
| ClinicalTrials.gov  | 30                  |

included 96 publications. The flowchart in **Figure 1** summarizes the review process.

**Supplemental File S2** (Supplemental Digital Content 2, <http://links.lww.com/PCC/B324>) is a table showing details of the 96 publications included in the final synthesis. Risk of bias assessment of each included study is summarized in a table in **Supplemental File S3** (Supplemental Digital Content 3, <http://links.lww.com/PCC/B325>). **Supplemental File S4** (Supplemental

Digital Content 4, <http://links.lww.com/PCC/B326>) summarizes all publications that reported on ECPR survival. **Supplemental File S5** (Supplemental Digital Content 5, <http://links.lww.com/PCC/B327>) summarizes all publications that reported on neurologic outcomes after ECPR. **Supplemental File S6** (Supplemental Digital Content 6, <http://links.lww.com/PCC/B328>) summarizes all publications that reported on or studied the relationship between CPR duration before ECMO cannulation and mortality.



**Figure 1.** Flowchart summarizing the results of the search and screening process: a total of 4,095 publications (3,870 + 225) were screened by three reviewers. Each publication was reviewed by at least two reviewers independently. Ninety-six publications were ultimately included in the final synthesis. ECPR = extracorporeal cardiopulmonary resuscitation.

### Definition of ECPR

There is no universally accepted definition of what constitutes ECPR. Extracorporeal Life Support Organization (ELSO) defines ECPR in its 2018 data dictionary as “the application of rapid deployment venoarterial ECMO, to provide circulatory support in patients in whom conventional CPR is unsuccessful in achieving sustained return of spontaneous circulation (ROSC). Sustained ROSC is deemed to have occurred when chest compressions are not required for 20 consecutive minutes and signs of circulation persist” (16). The 2019 Utstein reporting guidelines for IHCA aligned their definition of ECPR with this ELSO definition (17). Before 2018, the narrower ELSO definition of ECPR included only “ECLS used as part of the initial resuscitation from cardiac arrest”—patients who had achieved ROSC and were not receiving chest compressions (CCs) at the time of ECMO deployment did not meet the definition of ECPR (18). The 2015 Advanced Cardiovascular Life Support (ACLS) guidelines define ECPR as “venoarterial ECMO during cardiac arrest, including ECMO and cardiopulmonary bypass” (19).

The definition of ECPR used in published pediatric studies was not uniform—most studies used the pre-2018 ELSO definition (11, 20–28), but some considered ECPR to also include scenarios where patients underwent ECMO

cannulation shortly after CPR (i.e., after ROSC) (29, 30). The number of minutes post-ROSC by which cannulation must have occurred for an event to be defined as ECPR also varied in studies, with publications using cutoffs from 20 minutes up to 24 hours (29–31).

### Current Applications of ECPR in Pediatrics

The 2016 Pediatric ELSO Registry Report demonstrated increased utilization of both neonatal and pediatric ECPR from 2009 to 2015 (4). Almost all pediatric ECPRs in the literature are for IHCA. Patients with underlying congenital and/or acquired heart disease make up the majority of pediatric ECPR recipients (Supplemental File S2, Supplemental Digital Content 2, <http://links.lww.com/PCC/B324>).

Ortmann et al (9) reported that the frequency of ECPR use was 5.6% of 3,300 IHCAs reported to the Get With the Guidelines-Resuscitation Registry (GWTG-R) between 2000 and 2008, with the highest rate (19%) occurring in cardiac surgical patients. Matos et al (12) published similar numbers (6.7% and 20.7%, respectively) from the same registry during a similar time period. The incidence of ECPR use reported by Lowry et al (31) from the Kids' Inpatient Database was much lower—0.9% of 9,000 CPR events between 2000 and 2006. Alten et al (32) studied cardiac ICU (CICU) patients in the Pediatric Cardiac Critical Care Consortium registry between 2014 and 2016 and noted ECPR use in 27.2% of 485 cardiac arrests.

There is minimal literature on the use of in-hospital ECPR for pediatric out-of-hospital cardiac arrest, and this represents 2% of ECPR cases reported to ELSO (4). Likewise, although more common in adults (33–36), there is only one case report of out-of-hospital ECPR cannulation in pediatrics (37).

### Deployment and Resuscitation Practices

**Candidacy.** There were no studies of decision-making or candidacy for ECPR, including investigations on patient-specific factors. Few studies have evaluated ECPR logistics and the effects on outcomes, from which one can infer candidacy based on team-related factors (21, 27, 38). Inferring from patient outcome studies, those with noncardiac disease, lower pre-ECPR pH, and pre-ECPR renal injury have lower likelihood of survival (11, 30, 39).

**Timing of ECPR Activation and Impact of CPR Duration.** There is wide variation in practice regarding timing of the request for ECMO deployment during CPR, with centers describing ECMO activation within 1 minute of arrest and up to 20 minutes after CPR initiation, or after a specified number of rounds of resuscitative medications (20, 27, 29, 40–43). Data are conflicting about the relationship between CPR duration and mortality in pediatric ECPR (Supplemental File S6, Supplemental Digital Content 6, <http://links.lww.com/PCC/B328>). Plausibility supports that survival and neurologic outcomes will be worse with longer CPR duration before establishment of ECMO flows. Bembea et al (44), using merged data from GWTG-R and ELSO, found that median time to cannulation during ECPR was 48 minutes, and the relationship between

CPR duration and mortality was linear. Burke et al (38) found that median CPR duration before ECMO cannulation was significantly longer during nights/weekends compared with weekdays, and there was a higher rate of brain injury on imaging but no change in 1-year functional neurologic outcomes.

**Location of Deployment.** The 2016 Pediatric ELSO Registry Report described the location of cardiac arrest before ECMO cannulation in 1,828 ECPR cases between 2011 and 2015, revealing that the ICU was the most common location where cardiac arrest occurred (72%), followed by the operating room (OR) in 6%, ward in 5%, and emergency department (ED) in 3% (4). However, ECMO cannulation occurred more commonly in ICU (81%) and OR (8%) and less frequently in the ED (2%). These data suggest that some patients undergo intra-hospital transport, likely with CPR in progress, to an ICU or OR for ECMO cannulation.

Kane et al (29) demonstrated in a single-center cohort study that location of cardiac arrest was significantly different between survivors and nonsurvivors of ECPR in pediatric cardiac patients who suffered IHCA. Of their cohort of 172 patients, survival to hospital discharge was 49% when cardiac arrest occurred in the CICU ( $n = 129$ ), 73% when in the catheterization laboratory ( $n = 27$ ), and 35% when in other areas of the hospital such as ED, operating room, and ward ( $n = 16$ ) (29). Torres-Andres et al (41) found that location of ECPR was not different between survivors and nonsurvivors.

**Staffing and Logistics.** Published ECPR protocols are abundant in the literature, with staffing arrangements that ensure team availability and proximity (29, 40, 42, 43, 45). Ghez et al (21) demonstrated that their ECPR outcomes were comparable to outcomes in the literature despite not having an in-house rapid deployment ECMO team. Several protocols also ensure that a primed ECMO circuit is available at all times (29, 42, 43, 45). The 2016 pediatric ELSO report showed that 65% of ECPR cases had a circuit primed with blood products (4).

**CPR Quality.** Adult researchers were the first to demonstrate feasibility of measuring the quality of CPR (46, 47). Although several publications have added to our understanding of CPR quality in children receiving CCPR (48–52), there is a paucity of data regarding CPR quality in cases of pediatric ECPR. Taeb et al (53) performed a prospective single-center observational study comparing CPR quality between CCPR and ECPR by analyzing video recordings of resuscitations that were matched to telemetry and accelerometry data from each event. ECPR cases were found to have longer duration than CCPR, more frequent CC interruptions, lower end-tidal carbon dioxide, and CC rates which were less likely to be within goal range of 100–120 per minute. There is no published literature on the practice of fixed timed CC interruptions during pediatric ECPR, that is, CPR cycles with a minimum duration of compressions before allowing timed interruptions for cannulation to proceed.

**Vasoactive Administration During ECPR.** There are no consensus guidelines on vasoactive administration during ECPR (40). The impact of epinephrine dosing frequency and



total cumulative dose during CPR has been studied, with many studies demonstrating no difference in cumulative epinephrine dosing between survivors and nonsurvivors (6, 11, 29, 30, 54). In the first analysis of ECPR from GWTG-R, Raymond et al (11) reported that the total number of epinephrine doses was not significantly different between survivors and nonsurvivors. A follow-up propensity score analysis comparing CCPR to ECPR was published by Lasa et al (6) documenting higher cumulative epinephrine dosing in the ECPR population although longer CPR duration was also noted. The collinearity of total epinephrine dosing and CPR duration presents unique challenges to understand the physiologic impact of epinephrine on resuscitation practice and outcomes.

In a secondary analysis of the Therapeutic Hypothermia after Pediatric Cardiac Arrest (THAPCA) IHCA trial, Meert et al (30) described neurobehavioral outcomes 1 year after IHCA for the subset of children who received ECPR. Epinephrine dosing intervals in this cohort were variable, with epinephrine mostly administered at a longer dosing interval than the AHA-recommended 3–5 minutes. Ultimately, neither total doses of epinephrine nor dosing frequency were associated with outcomes.

**Cannulation Site and Strategy.** Significant variability in cannulation site strategy during ECPR is observed among centers (21, 22, 38, 42, 43, 45, 55–67). Multiple single-center reports reveal uniform site cannulation strategies (i.e., transthoracic cannulation only) (68–70) although the majority report a mix of peripheral and central strategies. Postoperative status, age, and weight are commonly reported as key factors determining cannulation site. Kane et al (29) analyzed the Boston Children's Hospital ECPR database between 1995 and 2008 and found no significant difference in survival to hospital discharge between patients cannulated peripherally and those cannulated transthoracically (29). Multiple other single-center and multicenter registry studies testing for associations between cannulation site and outcomes have similarly found no significant link with survival (24, 39, 41, 71, 72).

In contrast, some studies have shown an association between cannulation site and outcomes. Chan et al (73) studied ECPR recipients with cardiac disease using ELSO registry data and found a lower mortality risk for patients cannulated via the right carotid artery versus via the aorta. Meert et al (30) demonstrated that open-chest cardiac massage during ECPR was associated with higher 1-year survival and higher survival with good neurobehavioral outcomes, although it is unclear if the better outcomes were secondary to faster cannulation times, better CC quality, or simply the arterial cannula site. This study did demonstrate an association with shorter median duration from the start of CCs to the initiation of ECMO when cannulation was done via an open chest (30). Neurologic injuries after ECPR have been assessed using the ELSO registry and have been found to occur more commonly in patients cannulated peripherally (74), yet these findings are not consistent across specific subgroups (75, 76).

**Postcardiac Arrest Care.** There is no literature specific to postcardiac arrest care in ECPR. Outside of the ECPR population, there are general AHA guidelines (77, 78). There is no

literature specific to targeted temperature management after ECPR. The largest pediatric trials studying temperature management after cardiac arrest were the THAPCA trials, which showed that therapeutic hypothermia for comatose children after IHCA did not confer a significant survival benefit with favorable outcome at 1 year (79, 80).

**Neuromonitoring After ECPR.** There are no studies investigating neuromonitoring after ECPR. Outside of the ECPR population, research has focused on the routine use of electroencephalography (EEG) monitoring after cardiac arrest. Pediatric studies have shown that a higher EEG background score correlates with worse neurologic outcome and lower likelihood of survival (81–84).

## Outcomes

**Survival.** Lasa et al (6) analyzed 468 cardiac and 123 noncardiac pediatric patients from the GWTG-R registry who underwent greater than or equal to 10 minutes of CPR between 2000 and 2011 and found an increased odds of survival to hospital discharge for the ECPR patients compared with CCPR (adjusted odds ratio, 2.76; 95% CI, 2.08–3.65;  $p < 0.0001$ ) (6). Ortmann et al (9) also analyzed 3,323 pediatric IHCAs in the GWTG-R registry from 2000 to 2008 and found increased odds of survival for pediatric cardiac patients if they received ECPR compared with if they did not. Odds ratio for survival was 2.5 (95% CI, 1.3–4.5;  $p = 0.007$ ) in cardiac surgical patients and 3.8 (95% CI, 1.4–5.8;  $p = 0.011$ ) in cardiac medical patients (9).

The vast majority of ECPR literatures report survival outcomes, yet many studies are single-center analyses limited by small sample sizes and their retrospective nature (10, 20, 22–24, 28, 29, 41–43, 54–56, 58, 60, 68, 69, 71, 72, 74, 76, 85–88). The 2016 Pediatric ELSO Report includes the largest number of ECPR patients in the literature and reports survival for both neonates and pediatric patients to be 43% (4).

In a 2007 ELSO study by Thiagarajan et al (39), pediatric ECPR outcomes were reported. Majority of patients had cardiac disease, and survival to hospital discharge was 38%. Lower pre-ECPR pH and  $\text{Pao}_2$ , renal injury, pulmonary hemorrhage, CPR while on ECMO, and pH less than 7.2 while on ECMO were predictors of death. Raymond et al (11) found that the presence of cardiac disease was associated with survival in their ECPR study using GWTG-R registry. Supplemental File S4 (Supplemental Digital Content 4, <http://links.lww.com/PCC/B326>) summarizes publications that reported on ECPR survival.

**Neurologic Outcomes.** Neurologic outcomes in the pediatric ECPR population are most commonly reported for survivors at hospital discharge, with a paucity of reports on functional or neurobehavioral status beyond discharge. Most ECPR studies that report functional status use retrospectively assessed Pediatric Cerebral Performance Category (PCPC) and/or Pediatric Overall Performance Category scores (89). These are both six-point scales; lower scores representing higher performance. The assessment of neurologic function using these tools is subjective and not detailed. Additionally, the designation of a “favorable” outcome using these scores varies between studies, with some ECPR studies using a cutoff of less than or equal to 2 (mild

disability) (23, 28, 29, 38, 44, 45, 58, 72) and others using less than or equal to 3 (moderate disability) (6, 11, 12, 22, 27, 54). In the analysis by Bembea et al using merged data from GWTG-R and ELSO registries, discharge PCPC was only available in 116 of 241 ECPR survivors (48%). Ninety-three percent of those with available discharge PCPC data were considered to have a favorable neurologic outcome, with PCPC less than or equal to 2 (44).

Prospective assessments of neurobehavioral status in pediatric ECPR survivors have been done using the Vineland Adaptive Behavior Scales, Second Edition (VABS-II) tool, a neurodevelopmental tool that assesses the domains of communication, motor skills, socialization, and daily living (30, 90–92). The mean VABS-II score in the normative U.S. population is 100 (SD, 15), higher scores representing better function (93, 94). Using the THAPCA IHCA cohort, Meert et al (30) did 12-month follow-up assessments of 147 pediatric ECPR recipients, 51% of which were postcardiac surgery at the time of CPR. Sixty-one (41.5%) survived to 12 months postdischarge, 32 (22.1%) survived to 12 months with a VABS-II score decreased by less than or equal to 15 points from baseline, and 39 (30.5%) survived to 12 months with a VABS-II score greater than or equal to 70. Multivariable analyses revealed that open-chest cardiac massage was independently associated with better neurobehavioral outcomes. Of note, the open CCs group had shorter ECPR duration than the closed-chest group (37 vs 53 min;  $p < 0.001$ ) (30). In a separate study, Meert et al (92) showed that 12-month neurologic examination, cognitive outcomes, and neurobehavioral outcomes for ECPR recipients in the THAPCA IHCA cohort were no different from the patients who did not undergo ECPR.

Garcia Guerra et al (20) also prospectively evaluated neurologic outcomes in pediatric ECPR recipients. They assessed 17 of 25 ECPR survivors at 4.5 years old, and a minimum of 6 months after ECPR event. Mean Full-Scale Intelligence Quotient (FSIQ) score was 76.5 (SD, 15.9), which was lower than the U.S. population mean. Twelve of 17 subjects (70.6%) had FSIQ scores greater than 70, although seven were between 70 and 84, which represents borderline intellectual functioning. Only five of 17 (29.4%) had FSIQ greater than 84 (20).

Health-related quality of life in pediatric ECPR recipients has also been formally and prospectively assessed by Torres-Andres et al (41) using a validated tool—Pediatric Quality of Life Inventory. They found that those with normal brain imaging at the time of ECMO decannulation had higher quality of life scores (41).

As for reported neurologic complications, the 2016 pediatric ELSO report found that seizures developed in 6% of ECPR recipients. Cerebral infarct occurred in 5% of neonatal and 11% of pediatric ECPR patients. Intracranial hemorrhage was more common in neonates (14% vs 9%), and brain death was more common in pediatric patients (10% vs 2%) (4). There are no clinical investigations or guidelines on the determination of brain death while on ECMO, including in the ECPR population. Supplemental File S5 (Supplemental Digital Content 5, <http://links.lww.com/PCC/B327>) summarizes all publications that reported on neurologic outcomes after ECPR.

## Staff Training and Simulation

Several publications describe the use of simulation training for the performance of pediatric ECPR (40, 95, 96). These programs have resulted in improved processes and improved provider satisfaction. Other modes of staff training exist, with the goal of improving staff preparedness. Polimenakos et al (59) describe the use of preassigned nursing roles, and Tsukahara et al (72) describe the use of a flowchart.

## Cost-Effectiveness

Cost-utility analysis of salvage cardiac ECMO has been studied, but not in ECPR specifically (97). Lowry et al (31) published on costs of ECPR compared with conventional CPR (31), but the definition of ECPR used (ECMO within 24 hr of CPR) creates significant bias in the study results.

## DISCUSSION

We report a comprehensive systematic review of ECPR focusing on the pediatric population aimed at answering several key questions on this high-intensity and resource-heavy therapy. After a review of 4,095 publications, 96 studies were included in our analysis. There were only 40 publications specific to ECPR and 52 publications that included more than 10 ECPR patients in subgroup analyses. There were an additional four related publications that included relevant cross-sectional surveys. There are no randomized controlled trials studying ECPR in pediatric patients. The vast majority of studies on this subject are retrospective and single-center or registry-based, with the exceptions being few prospective long-term neurologic outcome studies.

The 2016 pediatric ELSO report demonstrated that from 2009 to 2015, neonatal ECPR use increased by 35% and pediatric ECPR increased by 67% (4). It is notable that most ECPR is deployed for cardiac patients. This is likely due to availability of staff who are trained in the rapid use of cardiopulmonary bypass, reversibility of the underlying process (e.g., arrhythmias), and possibly easier emergent application of ECMO in patients with an open sternum. In four separate publications, Ortmann et al (9), Morris et al (10), Raymond et al (11), and Matos et al (12) have shown that good outcomes after pediatric ECPR tend to favor cardiac patients.

There are several key clinical questions on ECPR that remain to be answered. Data on the relationship between ECPR duration and outcomes are conflicting—some studies show statistically significant associations between longer CPR duration and mortality (20, 22, 27, 28, 41, 43, 44, 54, 57, 58, 61, 68, 98), whereas others do not show an association (10–12, 23, 24, 29, 38, 42, 45, 55, 59, 62, 69, 71, 72, 86–88, 99, 100) (Supplemental File S6, Supplemental Digital Content 6, <http://links.lww.com/PCC/B328>). Cardiac surgical patients appear to be more likely to survive with good neurologic outcomes, even with prolonged CPR duration (9–12). Time to activation of ECMO also varies and is important because it affects the total CPR duration before achieving full ECMO support. Conversely, in the event that ECMO is activated too early, it may preclude

the resolution of reversible causes of cardiac arrest for which ROSC may have been achieved without ECMO.

The provision of AHA-recommended PALS therapies during ECPR has also come into question. In a survey by Lasa et al (40), only 64% of clinicians reported using prescribed dosing intervals for epinephrine during CPR before ECMO cannulation, whereas 19% of clinicians limited epinephrine doses. Limiting epinephrine doses theoretically prevents an increase in systemic vascular resistance that could affect myocardial recovery and ECMO flows after cannulation. Although this variation in practice is of uncertain clinical significance, it demonstrates the lack of standard practice during pediatric ECPR.

In considering ECPR utilization, one must consider: 1) its overall effectiveness compared with conventional CPR and 2) its cost-effectiveness. ECPR is typically deployed for CPR that has failed routine resuscitative therapies, and multiple studies have shown ECPR to be associated with improved survival when compared with CCPR in pediatrics (6, 10–12, 54). There is, however, a paucity of literature on the cost-effectiveness of pediatric ECPR. Additionally, no discussion of cost-effectiveness can be complete without a discussion about long-term functional and neurologic outcomes. Unfortunately, the neurologic outcomes literature is also wanting. Although there are many studies that have reported neurologic outcomes based on retrospectively collected PCPC data, only a few studies have followed pediatric ECPR survivors prospectively to assess outcomes. Additionally, several of the retrospective and prospective studies assessing neurologic outcomes are limited by missing data (6, 11, 20, 44). The few studies that have reported neurologic outcomes after prospective follow-up (20, 30, 41, 90–92) have potential confounding bias, since long-term neurologic and functional outcomes after ECPR will not only depend on the ECMO deployment process and ECMO complications thereto but also on prearrest care, postarrest care, underlying medical conditions, and subsequent medical events and treatment. Nevertheless, these prospective follow-up studies have added significantly to our understanding of neurologic outcomes after pediatric ECPR.

The lack of a universal definition for what constitutes ECPR means that some of the included studies in this synthesis may have patient populations that are vastly different from each other, presenting a challenge to cohesive interpretation. We did not perform a meta-analysis on the data due to study differences in ECPR definition, populations, and outcome measurements.

In reporting the outcomes from included studies, we are unable to account for patient management while on ECMO, including anticoagulation, mechanical ventilation, nutrition, infection, and sedation. We are also unable to account for prearrest management, CPR quality, and postresuscitation care.

## CONCLUSIONS

Current literature on ECPR use in pediatric patients is limited. Although ECPR use is on the rise and there are data to support its use, key questions remain surrounding ECPR candidacy, resuscitation practices before establishment of ECMO support,

neurologic outcomes, and cost-effectiveness. This report reviews the work which has been done thus far and highlights gaps in knowledge ripe for future research.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's website (<http://journals.lww.com/pccmjjournal>).

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